

Iliad 300 (S)TEM for materials science

The Thermo Scientific™ Iliad™ 300 (S)TEM is a fully integrated analytical (scanning) transmission electron microscope featuring the new Iliad EELS Spectrometer and Energy Filter with the dedicated Zebra EELS Detector, the new NanoPulser Electrostatic Beam Blanker, and a choice between the Thermo Scientific™ Dual-X or Super-X™ EDX Detection Systems.

The Iliad 300 (S)TEM is capable of operating at accelerating voltages between 30–300 kV. Thermo Scientific™ Velox™ Software facilitates access to all microscope modalities, while Thermo Scientific™ AutoScript™ Software enables advanced control of the TEM. The Iliad 300 (S)TEM, with its unprecedented hardware and software integration, allows for precise analytical investigation of advanced materials. It is also highly flexible, and a broad range of detectors and sample holders can be implemented.

Overall, the future-proof concept and state-of-the-art instrument design of the Iliad 300 (S)TEM combine to support your research at the frontiers of materials science.

Iliad 300 (S)TEM system energy resolution

Package	System energy resolution
EB: Extreme brightness cold field emission gun (X-CFEG)	0.4 eV
	0.3 eV at reduced extraction voltage
HR: High energy resolution	0.2 eV
UHR: Ultra high energy resolution	0.025 eV at 60 kV

Iliad 300 (S)TEM: STEM resolution and information limit

Package	STEM resolution (probe corrected)	STEM resolution at 30 kV (probe corrected)	Information limit (image corrected)
EB: Extreme brightness cold field emission gun (X-CFEG)	50 pm with 100 pA probe current	136 pm with 100 pA probe current	70 pm
HR: High energy resolution	50 pm with 30 pA probe current	125 pm with 30 pA probe current	60 pm
UHR: Ultra high energy resolution			

Note: All specifications are at 300 kV using an S-TWIN lens unless otherwise specified.

Key features

Integrated, advanced **EELS and EDX spectroscopy** with the Iliad EELS Spectrometer and Super-X or Dual-X EDX Detection System.

NanoPulser Beam Blanker enables dose optimization and time-resolved experiments.

Operating voltages from **30 to 300 kV** accommodate optimized experimental conditions for a broad range of materials and applications.

Velox Software facilitates data acquisition and processing.

AutoScript TEM Software offers advanced microscope control and customized workflows creation.

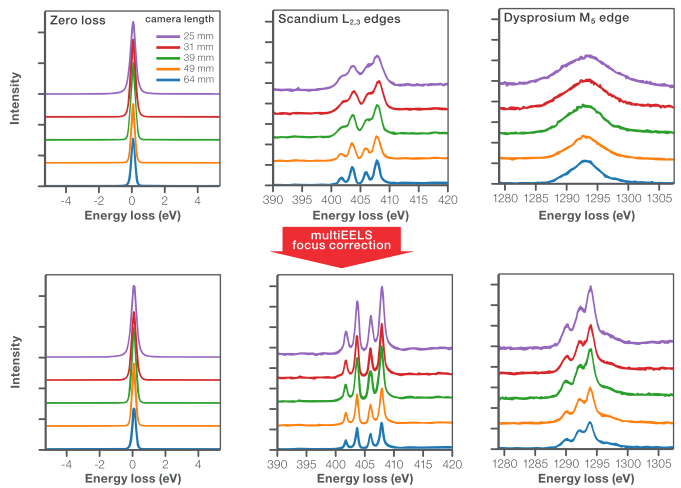
Flexible compatibility with a broad range of detectors and holders.



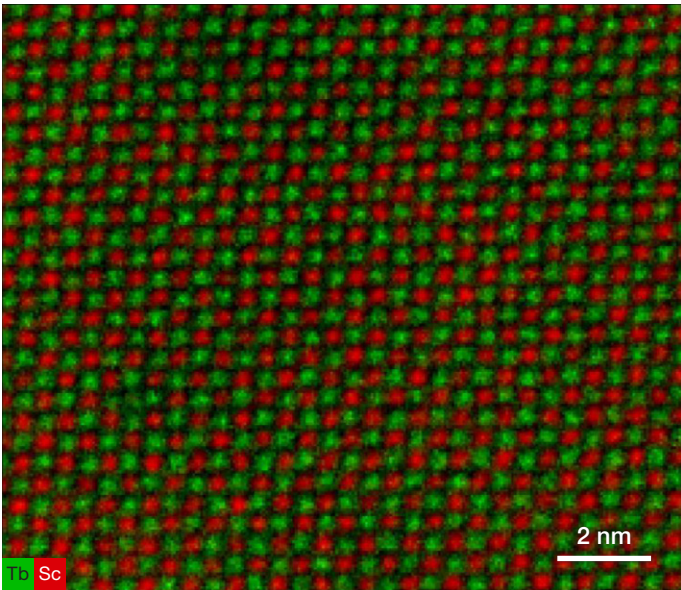
The Iliad 300 (S)TEM.

Iliad EELS Spectrometer: advanced optical integration for reliable materials analysis

In an optimal electron energy-loss (EELS) experiment, a broad range of electron energies (up to several thousand eV) must be simultaneously transferred through the microscope and spectrometer, from specimen to detector, without introducing any chromatic blur or distortions. At the same time, the TEM operator is continuously switching between different settings to maintain optimal conditions in both the microscope and spectrometer. For accurate transfer of the entire range of experimental conditions, close optical integration needs to be ensured; it is critical that the chromatic defocus in the microscope matches the focus of the spectrum. The Iliad (S)TEM addresses this challenge with its high stability and uniquely integrated optics, resulting in highly optimized EELS data collection.



MultiEELS Mode data collection for DyScO₃ at 60 kV, monochromated, with a 30 mrad semi-convergence angle, and 5 mm spectrometer entrance aperture. Data obtained at five different camera lengths. Top) Without correction. Bottom) With on-the-fly auto-correction for chromatic defocus. Data collected by Dr. Wouter Verhoeven.

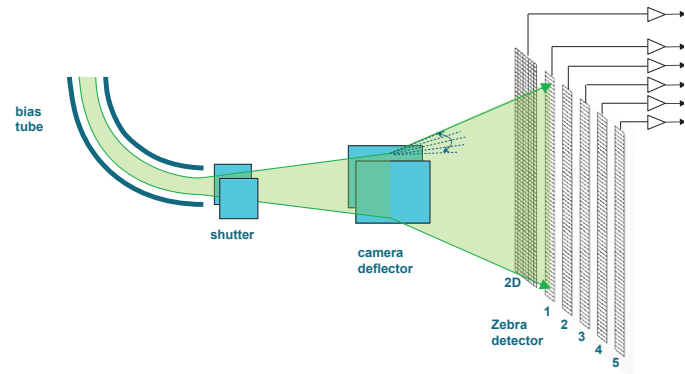


Atomically resolved EELS map of TbScO₃ with Tb shown in green and Sc shown in red. 1 eV per pixel dispersion was used. The 256x256 pixel map was recorded in 32 seconds. Data collected by Dr. Daen Jannis.

Iliad EELS Spectrometer	
Operating range (kV)	30–300
EELS mode	Yes
EFTEM mode	Optional
Energy range in single spectrum mode (eV)	>4000
MultiEELS Mode	Yes
Max non-isochromaticity at 200 kV (eV)	<0.2
Max chromatic distortion (% over 50 eV) at 300 kV	<0.1

Zebra Detector designed for EELS data collection

Dedicated for EELS data collection, the Thermo Scientific Zebra Detector consists of five independent 1D-strip detectors and one larger area designed for Iliad EELS Spectrometer alignment. The bias tube is used for fast electrostatic shifting of energy loss during MultiEELS data collection while the camera deflector provides fast electrostatic switching between the strips in the detector. This innovative multi-strip design allows for detection at a fast spectral rate, with readout speeds of up to 10,000 spectra per second. Thermo Scientific™ MultiEELS™ Mode enables the nearly simultaneous acquisition of up to 5 different energy ranges in the EELS spectrum under the exact same experimental conditions.



Schematics of the EELS-dedicated Zebra Detector.

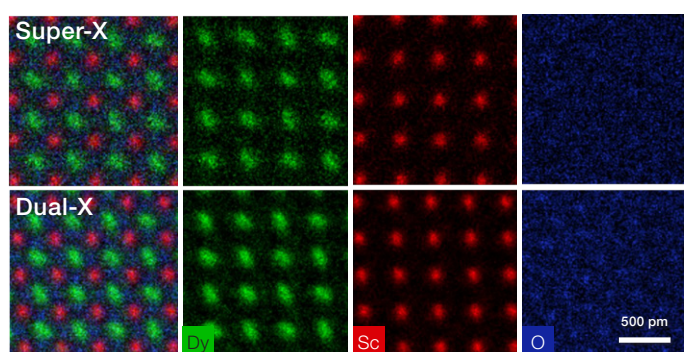
Zebra detector

Acquisition area	5 strips of 2048 pixels
Alignment area	1 alignment area of 256x2048 pixels
Pixel size	14 μm
Detector technology	Indirect 5 strip detector
Spectral rate (max. spectra/s)	10,000
MultiEELS Modes	1–5 EELS spectrum ranges

	Super-X Detector	Dual-X Detector
Output count rate	Up to 800 kcps	Up to 260 kcps
Energy resolution	≤136 eV for Mn-Kα and 10 kcps (output)	≤130 eV for Mn-Kα and 10 kcps (output)
	≤140 eV for Mn-Kα and 100 kcps (output)	≤140 eV for Mn-Kα and 100 kcps (output)
Solid angle (srad)	0.7	1.8
P/B ratio	Fiori number >4000	Fiori number >2000
Background signal	Below 1% Fe and Co spurious peaks in EDX system background	Below 2.5% Fe and Co spurious peaks in EDX system background

EDX detector options

The Iliad 300 (S)TEM comes with one of two EDX detector options that feature our symmetric, windowless design based on patented SDD technology; the Super-X or Dual-X Detection System.



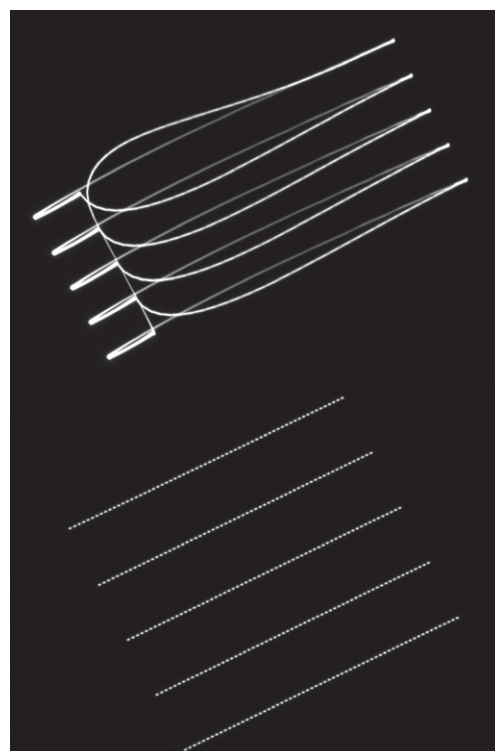
A quantitative comparison between Super-X and Dual-X Detectors for a DyScO_3 sample, using the same total electron dose. Sample courtesy of Dr. Lena Kourkoutis, Cornell University. Data collected by Dr. Cigdem Ozsoy-Keskinbora.

NanoPulser Electrostatic Beam Blanker

The Thermo Scientific NanoPulser Electrostatic Beam Blanker is used to pulse the electron beam and is placed between the electron gun and the condenser module. It blanks the beam by deflecting it away from the optical axis. The NanoPulser Beam Blanker enables nanosecond-scale control over the electron dose rate delivered to the sample, allowing for a wide range of applications from dose-efficient imaging to time-resolved experiments. Free running mode can be applied in both TEM and STEM, while the scan synchronized mode is aimed for STEM operation. In this synchronized mode, the NanoPulser Beam Blanker can make acquisitions up to 40% more dose efficient by eliminating scan overhead before the specimen.

NanoPulser Electrostatic Beam Blanker specifications

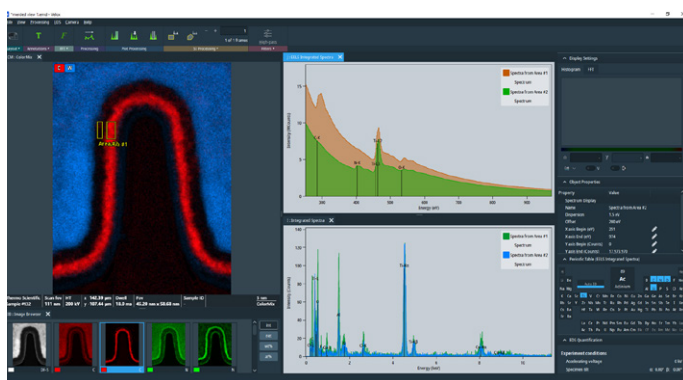
Various internal and external inputs are available to synchronize the scan engine, cameras, and free-running dose modulator.



Electron beam scanning over a sample with conventional (no blanking) STEM (top) and with a pulsating beam blanker (bottom), which provided 50% blanking during the scan while also completely blanking the flyback signal. Data collected by Dr. Noopur Jain.

Velox and AutoScript Software ecosystem for TEM

The Iliad 300 (S)TEM is powered by Velox Software, which offers cutting-edge, comprehensive experimental control. It facilitates access to STEM optics and detectors, enhancing reproducibility and yield, while also supporting quantitative STEM and TEM material analysis. Velox Software stands out with its integrated, easy-to-use, and ergonomic user interface, providing high-quality imaging and compositional mapping with both EDX and EELS, along with control of the NanoPulser Electrostatic Beam Blanker. Customized workflows often require advanced access to (S)TEM controls as well as the use of AI automation, which is enabled by Autoscript Software.



Velox Software user interface, featuring its intuitive and seamless design.

 Learn more at thermofisher.com/iliad300

thermoscientific

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